



Sensing the World

Look before you leap

움직이기 전에 주변을 먼저 살펴요

BIG PICTURE

A smart agent 🤖



It checks **ahead** and **below**, then decides what to do.

감지 → 결정 → 행동

HOW TO COUNT

Count the axis — one block, one number



- 1 Stand on the start block — that spot is $(0, 0, 0)$.
- 2 $x \rightarrow$ walk **right**. Add **+1** for every block.
- 3 $y \rightarrow$ jump **up**. Add **+1** for every block.
- 4 $z \rightarrow$ walk **forward**. Add **+1** for every block.

Say them in order every time: $(x, y, z) = \text{across, up, forward}$.

한 칸 움직이면 +1 — 오른쪽 x , 위로 y , 앞으로 z . 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft



^x
across →

^y
up ↑

^z
forward ↗

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press **C** in the world)
- ✓ Type the numbers in order — **(x, y, z)**
- ✓ Press **Run ▶** — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

agent.detect — look ahead

Ask the agent: is something **blocking the way**?

```
if agent.detect(DETECT_BLOCK, FORWARD):  
    agent.turn(LEFT_TURN)  
else:  
    agent.move(FORWARD, 1)
```

Wall ahead? Turn. Path open? Walk on.

앞에 막혀 있으면 돌고, 아니면 전진

CONCEPT

agent.inspect — what's below?

```
if agent.inspect(DOWN) == LAVA:  
    agent.place(STONE, DOWN)  
    agent.move(FORWARD, 1)  
else:  
    agent.move(FORWARD, 1)
```

Lava below? Cover it first, then move. Safe? Just move.

발밑이 용암이면 먼저 막아요

CONCEPT

Different block → different action



on DIRT

place a flower



on SAND

place a cactus

발밑 블록에 따라 다르게 행동해요

WATCH IT!

Check → branch → act

```
below = LAVA
```

```
below = GRASS
```

```
inspect(DOWN) == LAVA
```

```
✓ true → place(STONE, DOWN)
```

```
✗ false → move(FORWARD, 1)
```

```
▸ Action
```

조건에 따라 행동이 걸려요

PREDICT 🤔

What gets planted?

Predict first — then click reveal

```
# agent walks over: dirt, sand, dirt
for i in range(3):
    if agent.inspect(DOWN) == DIRT:
        agent.place(FLOWER, DOWN)
    if agent.inspect(DOWN) == SAND:
        agent.place(CACTUS, DOWN)
agent.move(FORWARD, 1)
```

Reveal output 🔍 reveal

VOCAB

New words today

detect check if a block is in a direction

inspect ask which block is there

condition a true/false test for **if**

else what to do when the test is false

감지하다

조사하다

조건

그 외에는



DEBUG

What's wrong with this code?

Goal: cover **lava below** — but the agent falls in.

buggy

```
if agent.detect(DETECT_BLOCK, FORWARD):  
    agent.place(STONE, DOWN)  
agent.move(FORWARD, 1)
```

The lava is below — where does this code look?



DEBUG

The bug

Bug: the check looks **FORWARD**, but the lava is **DOWN**. Inspect the block below.

fixed

```
if agent.inspect(DOWN) == LAVA:  
    agent.place(STONE, DOWN)  
agent.move(FORWARD, 1)
```

YOUR TURN!

Cross the danger field

☒ Check the block **DOWN** at least once

☒ Check the block **FORWARD** at least once

☒ Act differently for 2+ block types

☒ Cover lava before stepping on it

☒ Send a video of a safe crossing



QUIZ

QUESTION 1 OF 3

Lava is under the agent. Which check finds it?

```
player.say("lava")
```

```
agent.detect(DETECT_BLOCK, FORWARD)
```

```
agent.inspect(DOWN) == LAVA
```

```
agent.move(UP, 1)
```



QUIZ

QUESTION 2 OF 3

When does the `else` branch run?

Only on the first loop.

Never.

Always, every time.

Only when the `if` condition is false.



QUIZ

QUESTION 3 OF 3

Why check a block **before** moving?

It places blocks automatically.

It makes the agent move faster.

There's no reason to.

So the agent can decide what to do and stay safe.

QUIZ

Your score

0 / 3



Re-read the slides. You'll get it.

↺ retake



A sensing agent!

Next: the world reacts to you.

다음 주: 플레이어 이벤트